

PART-I

8)

Disturbance (road slope) is defined as step input starting at time zero. It is tried for two values: $\theta=7$ degrees and $\theta=-7$ degrees. In the below plot

Purple: 7 degrees road slope

Blue: No slope

Yellow: -7 degrees road slope

It can be seen that overshoot is decreased for the positive slope case. This is because now the system fights against gravity also, which adds additional damping. Contrary, for the case where the slope is negative the damping decreases and overshoot increases.

